

Module 06: Learning — Parameter Estimation, Bayesian Model Reduction, and Dirichlet Updates

Learning Objectives

1. Derive **parameter learning** equations: how the A and B matrices of a POMDP are updated from experience.
2. Derive **Bayesian Model Reduction (BMR)**: an efficient method for structure learning through post-hoc model comparison.
3. Formalize the **Dirichlet-Categorical conjugate model** and its update rules.

Introduction

Learning in Active Inference is the process of updating the generative model — both its parameters (connection strengths, likelihoods) and its structure (which variables to include). This module develops the mathematics of both parameter learning (updating sufficient statistics from experience) and structure learning via Bayesian Model Reduction.

Key Concepts

1. Parameter Learning in the POMDP

The A and B matrices of the POMDP are learned through experience. Using Dirichlet priors:

$p(A_{ij}) = \text{Dir}(A_{ij}; a_{ij})$ — each column of A has a Dirichlet prior with concentration parameters a_{ij}

After observing outcome o and state s , the update is:

$$a_{ij} \leftarrow a_{ij} + \eta \cdot o_i \cdot s_j$$

where η is a learning rate and $o_i \cdot s_j$ is the outer product encoding which outcome occurred in which state. This is the Dirichlet-Categorical conjugate update — the concentration parameters accumulate evidence.

Similarly for the B matrix:

$$\mathbf{b}_{ijk} \leftarrow \mathbf{b}_{ijk} + \eta \cdot \mathbf{s}_i(t) \cdot \mathbf{s}_j(t-1) \cdot \mathbf{u}_k(t-1)$$

where $\mathbf{u}_k$ is the action taken. This updates transition probabilities conditioned on action.

2. Bayesian Model Reduction

Bayesian Model Reduction (BMR) (Friston et al., 2017) enables structure learning without re-fitting the model from scratch. Given a full model M_1 with posterior parameters $\mathbf{a}$ (after learning), BMR evaluates a reduced model M_0 with modified prior parameters $\tilde{\mathbf{a}}_0$:

$$\ln p(\mathbf{o} | M_0) - \ln p(\mathbf{o} | M_1) \approx \ln B(\tilde{\mathbf{a}}_0) - \ln B(\mathbf{a}_0) - \ln B(\tilde{\mathbf{a}}) + \ln B(\mathbf{a})$$

where $B(\cdot)$ is the multivariate beta function and $\tilde{\mathbf{a}}$ are the posterior parameters under the reduced model:

$$\tilde{\mathbf{a}} = \mathbf{a} + \tilde{\mathbf{a}}_0 - \mathbf{a}_0$$

This provides an **analytic** model comparison without re-fitting — it computes the evidence for the reduced model from the learned parameters alone.

3. Dirichlet-Categorical Conjugacy

The Dirichlet distribution $\text{Dir}(\theta; \alpha_1, \dots, \alpha_K)$ is the conjugate prior for the Categorical likelihood $\text{Cat}(x; \theta_1, \dots, \theta_K)$:

$$p(\theta | \mathbf{x}_{1:N}) = \text{Dir}(\theta; \alpha_1 + \mathbf{n}_1, \dots, \alpha_K + \mathbf{n}_K)$$

where $n_k = \sum_i 1(x_i = k)$ is the count of observations in category k . Key properties:

- **Mean:** $E[\theta_k] = \alpha_k / \sum_j \alpha_j$
- **Variance:** decreases as the total count $\sum \alpha_k$ increases $\rightarrow$ more data = higher confidence
- **Entropy:** decreases with more evidence $\rightarrow$ progressive refinement of beliefs

Derivation Exercises

1. Derive the Dirichlet posterior update from a Categorical likelihood and Dirichlet prior using Bayes' rule.
2. Derive the BMR evidence ratio for comparing a full model (K parameters) with a reduced model ($K-1$ parameters).
3. Show that BMR with a "sleep" interpretation (evaluating model evidence during offline replay) provides a computationally efficient alternative to exhaustive model search.

Conclusion

Learning is inference over parameters and structure through conjugate Bayesian updating and Bayesian Model Reduction. Together with perception (state inference) and action (policy selection), this completes the core mathematics of Active Inference. Module 07 extends the framework to multi-agent settings.